Ziyi Xie

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https://github.com/messixieziyi/project-samples https://www.linkedin.com/in/ziyixienyu/

Skills: Python · SQL · PyTorch · Dask · NumPy · AWS · Snowflake · Machine Learning · Time Series Forecasting · Automation · Vibe Coding

Experience

Data Scientist @ Capital One — *McLean, VA* 2024 - Present

- Revamped valuation model infrastructure
  * Developed a new **NumPy- and Dask-based** calculation backend that improved runtime by 90% and reduced memory usage, enabling next-generation credit card decisioning.
  * Led migration of the valuation model onto the new infrastructure with deep understanding of the legacy system: drove **numerical consistency validation**, redesigned core components to be modular and extensible, and improved code efficiency and readability while correcting pre-existing errors discovered along the way.
  * Led end-to-end redesign of the **gaming** feature for the valuation system: surveyed legacy capabilities, authored a comprehensive **gap analysis report** with a prioritized improvement roadmap, secured senior leadership buy-in through executive presentations, and now progressively delivering enhancements with the team. The feature spans scenario and counterfactual analysis, sensitivity and stress testing, and strategy simulations across credit policy, pricing, segmentation (CL/APR assignment), line management, promotional offers, vintage gaming, portfolio mix, and acquisition channels.
- Led **0-to-1 product transformation** of internal cash flow calculation package (last step of all valuation).
  * Owned **project planning and execution** — translated an ambiguous problem statement into concrete tasks, defined the timeline, and coordinated work assignments across team members.
  * Worked with customers to understand **ambiguous uniqueness** in customer acquisition and customer management cashflow calculation, and eventually **simplified complexity** and consolidated multiple pipelines into one.
  * **Identified issues** with missing and problematic inputs, and developed a new feature to handle them automatically based on preset rules.

Data Scientist @ Amazon — *Seattle, WA and NYC* Summer 2021, 2022 – 2024

- Published **monthly forecast updates** for Amazon EC2 Server Demand for various AWS Data Centers over the next 2+ years. Collaborated with AWS teams to understand supply constraints and supply plan for EC2, and translate them into usable inputs for EC2 demand Forecast. Evaluated and published Peak Event (Prime Day, Black Friday, Cyber Monday) traffic, order rate, service throughput for scaling purposes starting six months prior and concluding with a final confirmation weeks before the event.
- Built customized **Time-series Forecast Validation** and **Anomaly Detection** mechanisms to automatically detect issues in forecast. Developed Automated Dashboards in QuickSight and Python scripts for Excel report automation, enhancing forecast visualization and snapshot-over-snapshot analysis. These reduced the duration of average human forecast review process to **1/3** of the original time.
- Enhanced algorithm of Amazon EC2 Server Demand forecast model by leveraging life cycle information of different generations of EC2 servers and optimize model structures, thereby improving the model accuracy by **15%** with a yearly impact of **$225 MM** and reducing daily forecast run time from 2 hour to 30 min .
- Used **data driven evidence** to support AWS Data Center migration as a result of imminent capacity constraints. Recommended strategy to migrate AWS services on Marketplace Level from one Large AWS region to multiple new AWS regions based on projected demand and supply.
- Prototyped analytical models to optimize financial planning strategies, enhanced forecasting accuracy by **31%** in Mean Average Percentage Error and **63%** in total budget allocation with **$ 1.26 B** impact. Designed metrics for forecast evaluation to quantify model performance in the scope of financial planning with various emphasis. This advancement accelerated the automation of financial planning at Amazon, laying the groundwork for the present automated financial planning model currently being used.
- Article on Amazon Science for the work from my team: **How CloudTune generates forecasts for the Amazon Store at Amazon Science Website** (linked or search for title) This article discusses how we generate forecasts for traffic and load on Amazon services.

Education

New York University 2020 – 2022

M.S. in Data Science *GPA:**3.9**/4.0*

- Past Projects(link:**https://github.com/messixieziyi/project-samples**) on the following topics: Transformer Model Analysis, Music Recommendation System, Gomoku Cognitive Modeling, Human Comment Generatoion, Default Risk Prediction

University of California, Santa Barbara 2016 – 2020

B.S. Honor Program in Physics, College of Creative Studies, Summer Research Fellowship *GPA:**3.9**/4.0*

- @ University of Tokyo, Bio-engineering Department, 2018 Summer : We built a lab setup of a laser imaging system with a high energy pulse laser and a laser-excited acoustic wave generation mechanism. Applied image processing techniques to analyze the deformation of cancer cells under mechanical force from acoustic waves using laser imaging.
- @ UC Santa Barbara, Mechanical Engineering Department, 2017-2019 : Designed and built silicon made dual-micro-channel of different size and shapes, and injected two streams of constant flows of liquid with different florescence through the channel under electromagnetic microscope. Used Particle Image Velocimetry software (based on MATLAB) to analyze liquid flow in channels of various shapes, identifying designs that enhance mixing, as evidenced by density variation.